

# Enhancement One Narrative

## 1. Briefly describe the artifact. What is it? When was it created?

This first enhancement for Software Design and Engineering is the biggest enhancement I am making to the project. This enhancement is where I am adding the ability for users to add new communities to the database. I had to create multiple new files and modify several existing files. However, this will set the groundwork for the other enhancements and the final project. I used AI again to help me with these additions through an iterative prompting process. I have just started to test the new functionality, but from the start there are multiple areas for improvement in the polish phase.

### Here are the new files created:

#### Backend:

- admin.js – this adds the routes for the admin account on the backend. This account is my account that is the only account that can authorize other user accounts. There is a new GET route to list all the users and a PATCH route to approve new user accounts
- user.js – this adds the routes for the user accounts on the backend. There are two routes for users. One POST route to register and one POST route to login.
- adminController.js – the controller for the new admin account. This file has functions to get the users and update user status.
- userController.js – the controller for the user accounts with functions to register and login.
- auth.js – provides security for the new accounts. Has functions for authentication, require approval for user accounts, and require admin to ensure only the admin account can approve user accounts.
- seed-admin.js – to seed the admin account in the database. Uses variables from the .env file (updated)

#### Frontend:

- AdminPanel.jsx (and .css) – This new component is the interface for the admin user to approve or reject user accounts.
- AuthModal.jsx (and .css) – this new component is how the users including admins login or new users can register

- AddCommunityModal.jsx (and .css) – this component allows users to add a new community. It is a form that has fields for all the required information. I need to do lots of testing with this form to improve the UX and ensure it validates inputs.
- AuthContext.jsx – has the context for the user that includes what actions they are authorized to do.

**Here are the modified files:**

**Backend:**

- package.json and package-lock.json – necessary dependencies for the new features.
- app.js – to wire in the new routes
- schema.prisma – added a model for users

**Frontend:**

- main.jsx – imported the AuthProvider from the new AuthContext.jsx file
- App.jsx – many updates to add the new components, accounts, and necessary authorizations
- api.js – added the functions loginUser, registerUser, createCommunity, fetchAdminUsers, updateUserStatus, and authHeaders.
- useCommunities.js – added functionality to refresh if a new community is added

**2. Justify the inclusion of the artifact in your ePortfolio. Why did you select this item? What specific components of the artifact showcase your skills and abilities in software development? How was the artifact improved?**

This entire project is the artifact. It demonstrates my ability to utilize AI tools to create new applications and/or functionality for customers. The specific artifacts that are being improved show the ability to create a custom interface for users to be able to login and update the database. Also added security features to authenticate and authorize the users.

**3. Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?**

I am not entirely sure. One of the outcomes I planned for this enhancement I know I have worked on. The “Develop a security mindset...” outcome is achieved through adding authentication and authorization via the admin and user accounts. All accounts use json web tokens (JWT) to provide security. The second outcome I listed is “Employ strategies for building collaborative...”. This outcome might be the one I struggle with the most. I feel this

outcome is difficult to quantify with a final project that I work on by myself. This would be an area where I could use more guidance.

Insights after finishing the project:

**Collaborating in a team environment** – added comments throughout to make the code easier to understand and maintain. Admin accounts make it easier to add other team members as the project grows.

**Communicating with stakeholders** – added the customer interface to make it easier for customers to interact with the website.

**Software engineering and database** – to make the customer interface I had to add multiple new modals to the front-end, routes to the back-end, and users to the database.

**Security** – user accounts use Json Web Tokens (JWT) for authentication. The admin account uses credentials that are hashed for security. Only admin approved user accounts can add new communities. The form to add new communities validates all inputs.

#### **4. Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?**

So far, I have mostly been using AI prompts to create the first iteration of the project and now to add functionality. I have spent some time testing this newest functionality and found multiple areas that need manual refactoring. I have already corrected a few issues from the AI code. For example, it had multiple files with the same name. They were in different packages, so technically correct, but I felt it might become confusing during development and testing. The UX for adding new communities is atrocious. The modal regularly closes by itself, and all the information is lost. It took me multiple tries to add a test community. Also, there is currently only minimal input validation. I also do not like that the AI added a login button to the main interface. I plan on changing it so that the user interface to login and add communities is separate. I would like to keep the main interface clutter free for the end users.

Ultimately, I am happy with the first draft of the added features for this enhancement. However, there is still plenty of room for polishing before I am willing to call it complete. I think this process of using AI tools to add new features in an iterative process is a skill set that will be valuable in the workplace, especially when coupled with a test to the requirements mindset. This process ensures that the customer's needs are met and minimizes bugs and security issues.